

Is There a Giant Cave Under the Moon? NASA Wants to Find Out

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There is a big hole on the Moon called the Marius Hill Pit. Scientists think it might lead to a huge cave or **lava tube** underground. A **lava tube** is a tunnel made by flowing lava long ago. NASA has picked a project called GIMLI to find out. Scientist Than Putzig of the Planetary Science Institute leads it. The tools will ride to the Moon on a **lander** and a **rover**.

GIMLI will use **radar**, sensors that feel vibrations in rock, a tool that measures gravity, and cameras. Together, they could show if an empty space stretches away from the pit and how big it is.

A cave could someday protect astronauts from **radiation** and from the Moon's very hot and very cold temperatures. It could also hold clues about the Moon's **volcanoes** long ago.

Even if there is no cave, the mission could help explain how the pit formed.

Word Treasure

lava tube -- A lava tube is a tunnel left behind by flowing lava.

lander -- A lander is a spacecraft that sets down gently on the surface of a moon or planet.

rover -- A rover is a robot car that drives around to explore another world.

radar -- Radar sends out signals and listens for echoes to find things that are hard to see.

radiation -- Radiation is invisible energy from space that can be harmful.

volcanoes -- Volcanoes are openings where hot melted rock comes up from deep inside.

Background you need

Long ago, hot lava flowed on the Moon, and some of it may have left empty tunnels called lava tubes.

The Moon has no air to block harmful radiation, and it gets very hot and very cold.

Scientists send landers and rovers to study places that people cannot easily reach.

Break it down

WHO: NASA and scientist Than Putzig of the Planetary Science Institute

WHAT: A project called GIMLI will check whether a Moon pit leads to a giant cave

WHERE: The Marius Hill Pit on the Moon

WHY: A cave could protect future astronauts and hold clues about the Moon's old volcanoes

Why it matters

A safe cave on the Moon could someday help astronauts live and explore there.

Test what you remember

1. What is the name of the big hole on the Moon?

- ☐ (A) The Big Dipper
- ☐ (B) The GIMLI Hole
- ☐ (C) The Marius Hill Pit
- ☐ (D) The Apollo Crater

2. What is a lava tube?

- ☐ (A) A tunnel made by flowing lava
- ☐ (B) A tool for measuring gravity
- ☐ (C) A kind of rocket
- ☐ (D) A space toy

3. What is the project called?

- ☐ (A) GIMLI
- ☐ (B) APOLLO
- ☐ (C) GEMINI
- ☐ (D) ARTEMIS

4. How will the tools get to the Moon?

- ☐ (A) In a hot air balloon
- ☐ (B) On a submarine
- ☐ (C) On a lander and a rover
- ☐ (D) By an astronaut walking

5. How could a cave help astronauts someday?

- ☐ (A) It could make the Moon brighter
- ☐ (B) It could help them fly
- ☐ (C) It could protect them from radiation and very hot and cold temperatures
- ☐ (D) It could give them pizza

6. What could the mission learn even if there is no cave?

- ☐ (A) How to grow plants
- ☐ (B) Why the sky is blue
- ☐ (C) What time it is on Mars
- ☐ (D) How the pit formed

Answer key (try the quiz first!): 1: C 2: A 3: A 4: C 5: C 6: D

MY THOUGHTS (at least 20 words):